2023 IEEE International Symposium on High-Performance Computer Architecture (HPCA) | 978-1-6654-7652-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/HPCA56546.2023.10071066

A Systematic Study of DDR4 DRAM Faults in the Field

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***Abstract*—This paper presents a study of DDR4 DRAM faults in a large fleet of commodity servers, covering several billion memory device-hours of data. The goal of this study is to understand faults in DDR4 DRAM devices to measure the efficacy of existing hardware resilience techniques and aid in designing more resilient systems for future large-scale systems.**

The study has several key findings about the fault characteristics of DDR4 DRAMs and adds several novel insights about system reliability to the existing literature. Specifically, the data show sixteen unique fault modes in the DDR4 DRAM under study, including several that have not been previously reported. Over 45% of the faults that occurred affected multiple DRAM bits. The time-to-failure characteristics of faults internal to the DRAM die differ from those external to the DRAM die. We also examine faults from multiple DRAM vendors, finding that fault rates vary by more than 1.34x among vendors.

Finally, we use the data to compare chipkill ECC and an ECC that covers a DDR5 "bounded fault." Given the fault rates in this data, a bounded fault ECC increases the rate of faults that cause uncorrectable errors by up to 5.71 FIT per DRAM device compared to chipkill ECC.

I. Introduction

As data centers computation requirements grow, the total memory capacity provisioned is expected to grow as well. Several publications have shown that DRAM faults are a common source of failures in the field [1] [2] [3] [4] [5] [6]. Most of this memory is comprised of dynamic random-access memory (DRAM). Therefore, with the increase in memory capacity demands, the impact of DRAM faults on system reliability may become more severe, and future generations of systems and data centers may require significant increases in the reliability of memory subsystems.

One key to designing a reliable system is to understand the nature of faults that can occur. One way to obtain this understanding is by studying faults that do occur in current production environments. Insights obtained from such a study can be used by system designers and operators to measure the effectiveness of existing hardware resilience techniques and develop new mitigation schemes for future systems.

This paper which builds upon prior DRAM field studies [1] [2] [3] [4] [5] [6] [7], presents a large-scale field study of DDR4 DRAM errors from thousands of server nodes in production across multiple data centers at Google. The data covers several billion DRAM device-hours across 270 calendar days. The analysis uses errors logged by the servers to identify fault modes and attribute these fault modes to specific components of the memory subsystem (e.g., a DRAM row or bank).

Additionally, the paper models the reliability of future DRAM subsystems. Specifically, the DDR5 DRAM specification introduces the concept of a bounded fault (BF) [8, 9]. The intent of this concept is to provide guarantees on how DRAM internal components (such as a row) are mapped to DRAM data pins. One consequence of this guaranteed mapping is that system designers can implement error correcting codes (ECCs) that cover a subset of DRAM device data pins and be certain that specific fault modes remain correctable. This paper uses the DDR4 data to model the reliability impact of several such alternatives.

There have been several studies examining DRAM reliability in production data centers dating back many years, largely examining older memory technologies such as DDR1, DDR2, and DDR3 [1] [2] [3] [4] [5] [6]. While some public studies are available about DDR4 DRAM faults [7, 10], our paper contains analyses not performed in these two papers, including: the effects of DRAM vendor choice on DRAM faults; sixteen unique fault modes, the rate of transient and permanent DRAM faults and the time-to-fail analysis of components that are internal or external to a DRAM device.

This study has several key findings:

- Overall fault rates vary among DRAM vendors in our study by 1.34x. Transient fault rates vary by

1.07x; intermittent fault rates vary by 1.45x; and permanent fault rates vary by 2.04x.

- ~45% of the faults that occur in the DDR4 DRAM system under study affect multiple bits, such as faults that affect all or part of a DRAM row, column, or bank. This indicates that multi-bit faults occurred almost as frequently as single-bit faults in the DDR4 DRAM subsystems under study, a marked change from prior DDR DRAM technologies [2] [11].
- The data show several multi-bit fault modes that have not previously been reported, including fault modes that affect a quarter or half the banks in a DRAM device, and fault modes that affect multiple columns or rows within a single DRAM bank.
- Faults that can be attributed to components internal to a DRAM device (e.g., a row or a bank) have different time-to-fail characteristics than faults attributed to components external to the DRAM device (e.g., a pin or a connector), potentially implying different underlying fault types.
- Projecting ahead to DDR5 DRAM, and assuming the same distribution of fault modes as seen in this study, an ECC that only corrects bounded faults may increase the rate of faults that cause uncorrectable errors in a single DRAM device by 5.71 FIT per DRAM device compared to an ECC that corrects all faults from a single DRAM device (commonly known as chipkill).

The rest of this paper is organized as follows. Section II defines the terminology we use in this paper. Section III discusses related studies. In Section IV, we provide some background. Section V discusses the configuration of the nodes. Section VI discusses our methodology. Section VII presents the analysis of DRAM, DIMM, and channel faults. In Section VIII, we compare three ECC schemes: SEC-DED, Chipkill, and bounded fault. Finally, Section IX concludes our paper.

II. Terminology

This paper distinguishes between a fault and an error as follows [2, 12]:

- A *fault* refers to the underlying cause of an error, such as a stuck-at bit or high-energy particle.
- An *error* is an incorrect state resulting from a fault, such as an incorrect value returned from memory to the processor.

In other words, faults are an indicator of memory reliability while errors are an indicator of the impact of a fault on the rest of the system. Note that an error requires both that a memory location is faulty, and that the expected value of the memory is different than the value produced by the fault. For instance, a 0-value written to and read from a location with a stuck-at-0 fault will not return an error, while a 1 value written to that same location will return an error.

A fault's *type* is typically classified as *transient*, *intermittent*, or *permanent* [2, 12]. Transient faults are random, non-repeatable faults, and are expected in normal operation: they do not indicate damaged hardware. Intermittent and permanent faults repeatably occur in the same location. Intermittent faults only occur under certain conditions, while permanent faults occur every time a location is accessed.

A fault's *mode* is defined as the portion of the device or subsystem that is affected by the fault. For instance, a single-row fault mode is one that affects bitcells within a DRAM row, but no bitcells outside of that row.

The DRAM errors discussed in this paper are classified as either corrected or uncorrected [2, 12]:

- A *corrected error* is an error where the node hardware has sufficient redundant information (e.g., ECC bits) to recreate the intended value of the data and return this value to the system. This type of error does not affect the correct operation of the node, though it may result in a temporary or permanent performance reduction.
- An *uncorrected error* is an error where the node hardware has insufficient information to recreate the intended value of the data. This type of error can result in the termination of the process trying to consume that data or in a crash of the node, depending on the specifics of the error and the resilience features available on the node.

III. Related Work

There have been several studies examining faults and errors in DRAM in production systems over several years. Schroeder and Gibson presented a study of DRAM failures in supercomputer systems at Los Alamos National Labs (LANL) [3]. Schroeder et al. analyzed DDR1 and DDR2 DRAM errors from Google's server fleet [6]. Li et al. published a study of DDR2 DRAM errors on three different data sets, including a server farm of an internet service provider in 2007 and another expanded study of DDR2 DRAM errors in 2010 [13] [14]. Hwang et al. presented an expanded study of DDR1 and DDR2 DRAM errors on Google's server fleet and two IBM Blue Gene clusters [15].

More recent studies have included detailed breakdowns of fault modes and rates. Sridharan and Liberty presented a study of DDR2 DRAM failures in a high-performance computing system [2]. Sridharan et al. published a study of DDR3 DRAM faults with a focus on positional and vendor effects in 2013 [11]. Sridharan et al. presented a DDR3 DRAM study with a focus on the reliability impact of hardware resilience schemes from high-performance computing systems in 2015 [1]. Siddiqua et al. published a study of DRAM failures from client and server systems [16]. Meza et al. analyzed the DDR3 DRAM errors from

Facebook's server fleet and showed that errors follow a pareto distribution with a decreasing hazard rate [5]. Recently, Chen et al. present a study of DDR3 DRAM errors collected from one of Alibaba's data centers [4]. Gurumurthi noted a substantial increase in DDR4 fault rates relative to DDR3 based on data from a large-scale production data center [10]. Recently, Ferreira et al. presented a study of DDR4 DRAM errors in large-scale HPC Arm systems in 2022 [7].

There have been laboratory testing on DRAM dating back several decades (e.g., [17] [18] [19] [20] [21]), as well as studies on DDR3 and DDR4 DRAM disturbance faults [22, 23, 24]. Lab studies such as these allow an understanding of fault modes and root causes, which complement field studies that identify fault modes that occur in practice.

Finally, Criss et al. present an overview of the DDR5 "bounded fault" concept and state that it can be used by system designers to reduce ECC overheads [8]. However, the study did not provide any quantitative analysis of such a tradeoff.

IV. Background

DRAM subsystems are typically protected by multiple reliability features. This section provides a brief overview of techniques relevant to understanding this study.

An *error correcting code (ECC)* stores several additional check bits along with each data word [2] [25]. ECC detection and correction are performed on every read access to DRAM. The ECC can detect and/or correct certain errors that occur in the data and check bits. The errors that can be corrected depend on the specific ECC used by the hardware. There are two different ECCs of relevance to this work. A Single Error Correct-Double Error Detect (SEC-DED) ECC can correct any single-bit error and detect any double-bit error occurring in the data or check bits. By contrast, a "chipkill" ECC can correct any error confined to a single DRAM device [2] [10] [26] [27].

A *patrol scrubber* periodically reads every location in memory [2] [28]. If the read results in a corrected error, the patrol scrubber writes back the corrected data to memory. The goal of a patrol scrubber is to find DRAM locations with a transient fault and correct them before a second fault occurs, potentially causing an uncorrectable error. The time that the patrol scrubber takes to cycle through every location in a node's DRAM memory is called a scrub interval.

A *redirect scrubber* [29] writes corrected data back to DRAM after a corrected error is detected on a demand read (e.g., a read from the running program). The goal of a redirect scrubber is to immediately correct data in active use to avoid repeated corrected errors or future uncorrected errors.

V. Node Configuration

We collected data from tens of thousands of server nodes in production across multiple data centers at Google. Each node in the study contains two 64-core AMD EPYC™ 7002 processors. Each processor has eight memory channels. Each memory channel is populated with two 32GB DDR4 RDIMMs [30] [31], for a total of 64GB of DRAM per memory channel. The server nodes contain DRAMs from two different memory vendors; each node uses DIMMs from a single vendor. Nodes in this study were put into deployment with 'fresh' (new) DIMMs. DIMMs from vendors A and B were co-located in the same data centers and were targeted by a similar set of workloads.

A channel is comprised of multiple data wires (DQs) that each connects the processor to the DIMMs. A group of 4 DQ wires and their associated strobe (DQS), or source-synchronous clock, is called a lane. A memory channel has 18 lanes, each of which connects to both DIMMs on the channel.

Each DIMM consists of two ranks of 18 DDR4 DRAM devices each with four data (DQ) signals (known as an x4 DRAM device). Each rank is accessed independently. A single DRAM fetch accesses all 18 DRAM devices on a rank. Sixteen of the DRAM devices on a rank are used to store data bits and two are used to store check bits.

Each DRAM device contains sixteen internal banks that can be accessed in parallel. Logically, each bank is organized into rows and columns. Each row/column address pair identifies a unique 4-bit word in the DRAM device.

The nodes in this study use a chipkill ECC. Each node has a patrol scrubber configured to access all of DRAM once per 24 hours and a redirect scrubber.

VI. Methodology

A. Data Collection

The processors are configured to log DRAM errors into x86 Machine Check Architecture (MCA) registers [32]. Each node is configured to record any events logged in the MCA to the system event log. The event logs contain sufficient information to identify the specific faulty DRAM channel, rank, device, bank, row, column, and bit(s) for logged corrected errors. These event logs were retrieved from nodes running a variety of production workloads in the data center and analyzed using a set of post-hoc analysis tools.

This study does not collect DIMM temperature or other environmental conditions. Hence, the study cannot examine the effects of these variables on fault characteristics. These variables have all been shown to affect DRAM fault rates [10] [11] [33]. Therefore, other data centers, node configurations, or environments may experience different fault rates [1] [5] [11] [12]. However, experience shows that DRAM fault modes are similar regardless of environment [1] [5]. Additionally, information about workloads running on specific nodes is not available. While this information can be valuable in performing root cause analysis, the goal of our study is not to root cause memory faults, but to present the types of faults that systems may experience in the field.

B. Classifying Fault Modes

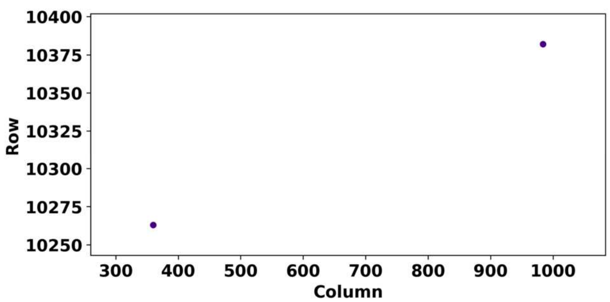


Figure 1. Some DRAM devices had errors on exactly two bits within a bank. Based on the rate of occurrence, these were classified as two single-bit faults rather than a single two-bit fault.

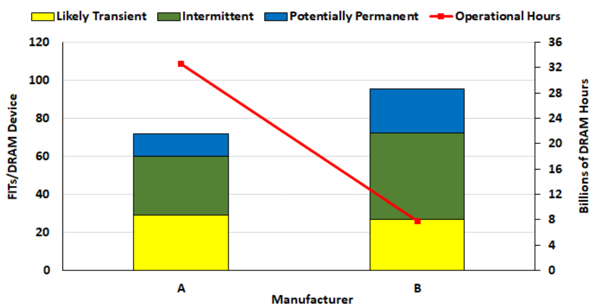


Figure 2. Fault rates (left axis) and operational hours (right axis) by Vendor.

Examining multiple errors logged on a single node yields a "map" of faulty bits in that node's DRAM. Making the simplifying assumption that all errors from a single DRAM device are from a single fault, one can use this map to identify a fault mode. This assumption has been used by prior studies [1] [11].

Given the scale of this study and the observed rate of faults, however, this simplifying assumption is not always appropriate – that is, a non-trivial number of DRAM devices are expected to experience more than one fault during the study. For example, Fig. 1 shows an error pattern that occurred within a single DRAM bank of a DRAM device in the study. The errors are confined to just two bits. Therefore, this could indicate either two independent single-bit faults or a single two-bit fault.

To properly classify this two-bits-in-a-bank error pattern, we use the following methodology to determine if we can reject the hypothesis that these are two independent faults. First, we identify the total rate at which this error pattern occurs, denoted by T_1. We then identify the rate at which two single-bit faults occur in different banks of the same device, denoted by T_2. If the hypothesis is correct, T_2 should be approximately equal to $15 * T_1$ given that there are 16 banks per DRAM device. The data in fact show that $T_2 \approx 15 * T_1$. This means that this error pattern occurred as often as would be expected for two independent faults. Thus, we cannot reject the hypothesis that the two-bits-in-a-bank error pattern is two independent single-bit faults, and therefore we classify this error pattern as two independent single-bit faults.

All fault modes described in subsequent sections of this paper occurred often enough to reject the hypothesis of two independent faults; that is, T_1 is larger than the appropriate function of T_2. Thus, all fault modes described in this paper are considered unique.

C. *Classifying Fault Types*

After classifying a fault's mode, we classify its type – transient, intermittent, or permanent.

For multi-bit fault modes that span more than one fetch from DRAM (e.g., multiple cache lines), we classify a fault as likely transient if all errors from that fault occur within a 24-hour interval (i.e., one patrol scrub interval). We cannot conclusively say the fault is transient because we cannot rule out that the fault will produce errors after the study period; however, prior studies have shown that this is the likeliest conclusion [2]. Conversely, if at least one error from the fault is logged in every 24-hour interval in the study period, we classify that fault as potentially permanent. Finally, if at least one 24-hour interval in the study does not have an error from the fault, we classify the fault as intermittent.

Due to the presence of a redirect scrubber, we use a different methodology to determine the fault type for single-bit faults that are confined to a single DRAM fetch. If only one error is logged from that fault, we classify the fault as a likely transient. If more than one single-bit error occurs, we distinguish potentially permanent faults from intermittent faults based on the approach discussed above. Note that the redirect scrubber is not instantaneous, so, we cannot rule out two errors that may be logged as the result of a single transient fault. However, this will be rare given the short average latency of the redirect scrubber relative to the error logging process.

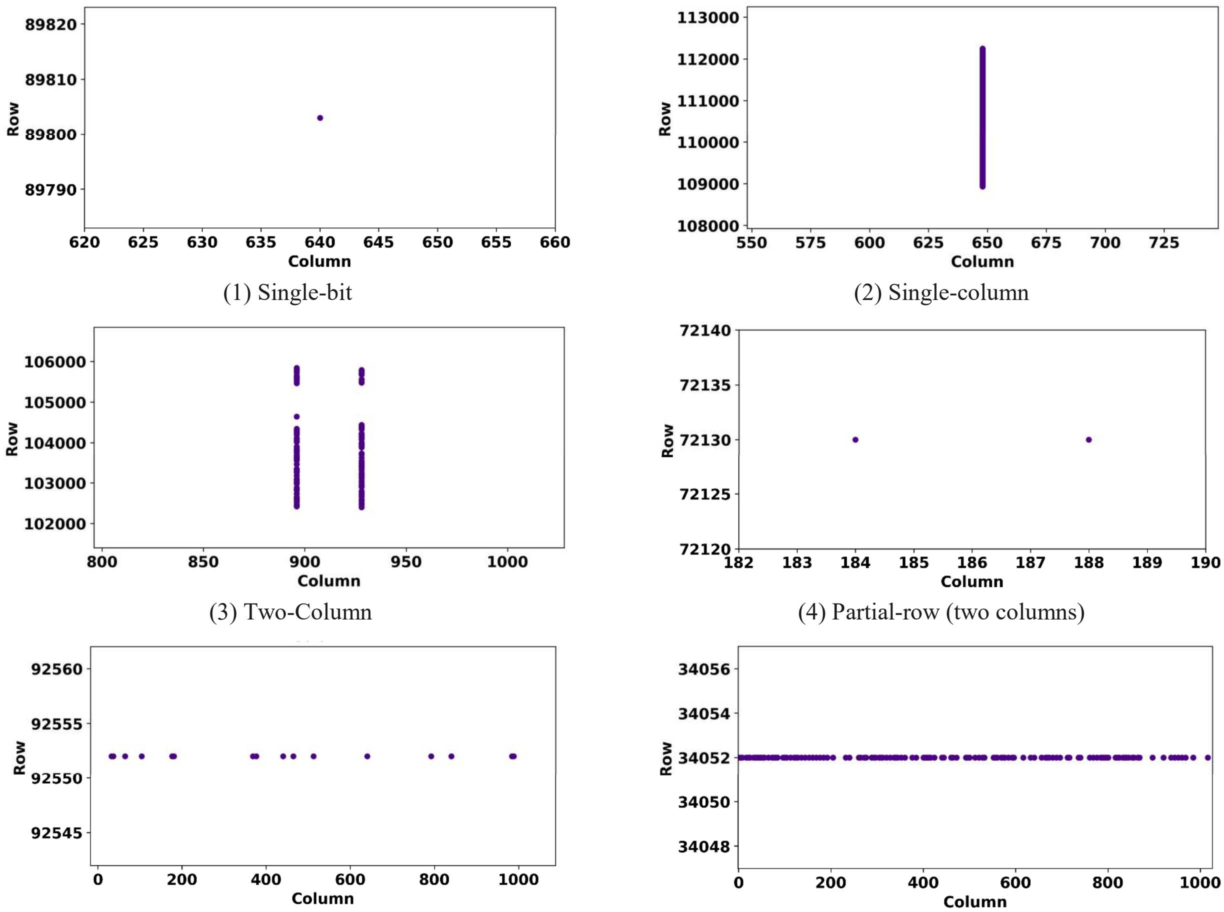
D. *Determining Fault Rates*

We are interested in knowing the rate at which each fault mode occurs. To quantify fault rate, we use a metric called *FIT per DRAM device* that indicates the rate of faults that occur in one billion device-hours of operation. We calculate an upper bound for FIT per DRAM device by:

$$FIT\ per\ DRAM\ device = \frac{Gamma^{-1}_{F+\frac{1}{3},1}(0.6)}{D \times T}$$

Where F is the number of faults, D is the number of DRAM devices and T is the total device-hour in billion hours. We use $Gamma^{-1}_{F+\frac{1}{3},1}(0.6)$ as in equation (3) in [34] to provide the 60% confidence upper bound. Using this formula, based on the Kerman neutral prior [35], improves the accuracy of the confidence bound for fault modes with low observed failure rates (e.g., zero maximum likelihood estimate FIT).

We also plot time-to-fail (TTF) for each fault of a given mode; this provides the inverse of the fault rate for that mode.



(1) Single-bit
(2) Single-column
(3) Two-Column
(4) Partial-row (two columns)
(5) Partial-row (sixteen columns)
(6) Single-row (146 columns)

Figure 3. Unique fault modes were observed during this study (part 1).

To determine time-to-fail for a fault, we must know the length of time between a node's deployment and the fault's first occurrence on that node. While we cannot precisely know a fault's first occurrence, we can know when a fault was first accessed because the node logs an error. Therefore, we define TTF as the length of time from a node's first deployment ($T_{in\text{-}service}$) to the fault's first logged error ($T_{first\text{-}fail}$), or $T_{first\text{-}fail}$ - $T_{in\text{-}service}$, and use this as a heuristic for the fault's time to first failure.

On all TTF plots, $T_{in\text{-}service}$ is plotted as day 0 and 100% represents the total number of faults that occurred within the nodes' first 180 days of service.

VII. DRAM, DIMM, and Channel Faults

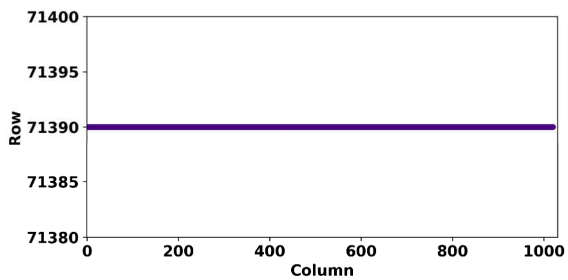
In this section, we present the DRAM fault rates, fault modes, and fault types that occurred on the nodes in our study.

A. *Fault Rates by Vendor*

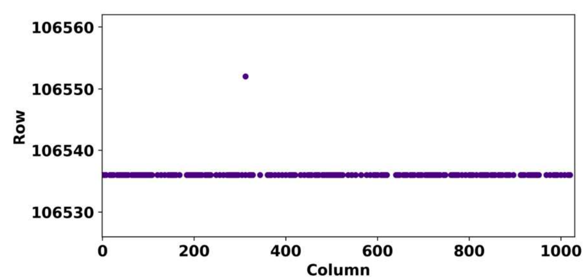
Fig. 2 (i.e., red line using the right y-axis) shows the aggregate number of DIMM-hours per DRAM vendor. Our observation period consists of 32.6 and 7.8 billion device-hours for DRAM vendors A and B, respectively. Therefore, we have enough operational hours on each vendor to make statistically meaningful measurements of each vendor's fault rate. Fig. 2 (i.e., stacked bar using the left y-axis) shows the fault rate experienced by each vendor during this period, divided into likely transient, intermittent, and potentially permanent faults. The figure shows a difference in fault rates between vendors. Vendor B has a 1.34x higher fault rate than Vendor A. This figure also shows that Vendor A has a 1.07x higher transient fault rate than Vendor B, while Vendor B has a higher intermittent and potentially permanent fault rate than Vendor A by 1.45x and 2.04x, respectively.

As results show, the fault rates obtained in our study are lower than those reported by Gurumurthi [10]. The difference in fault rate may be due to 1) qualification tests and DIMM screening performed by Google to identify and reject parts that do not perform well and 2) different DIMM part numbers or DRAM vendors, both of which have been shown to affect fault rates [11].

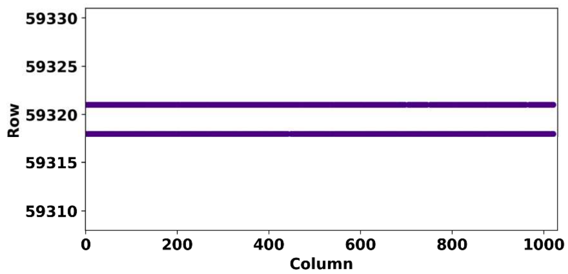
B. *Fault Modes*



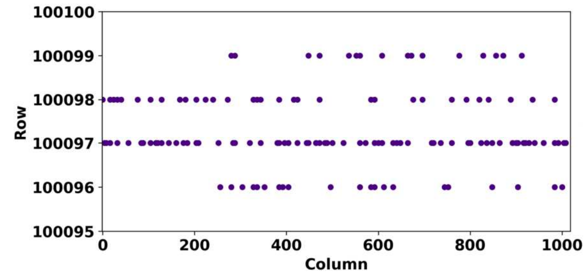
(1) Single-row (1024 columns)



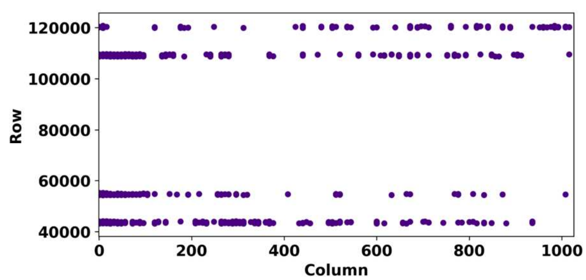
(2) Single-row-plus-single-bit



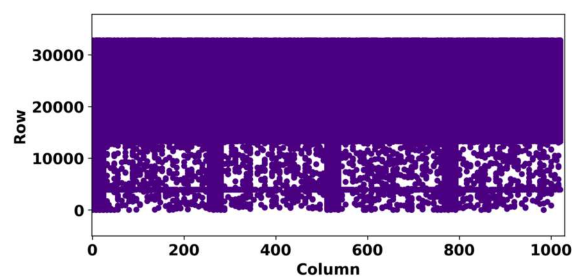
(3) Two-row



(4) Consecutive-row



(5) Cluster-row



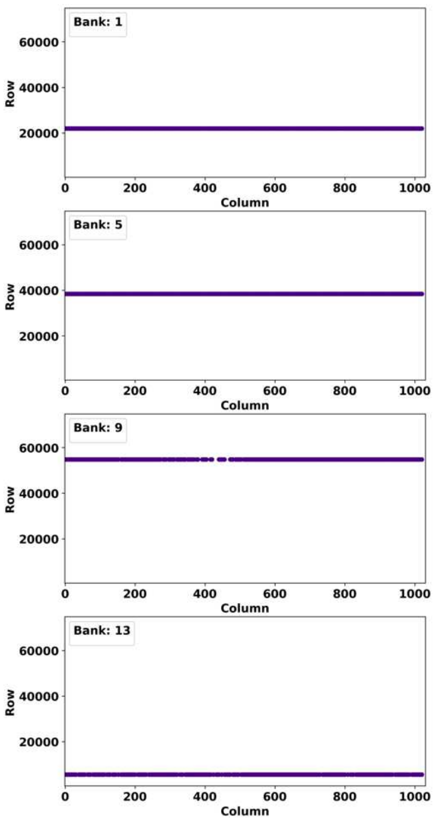
(6) Single-bank

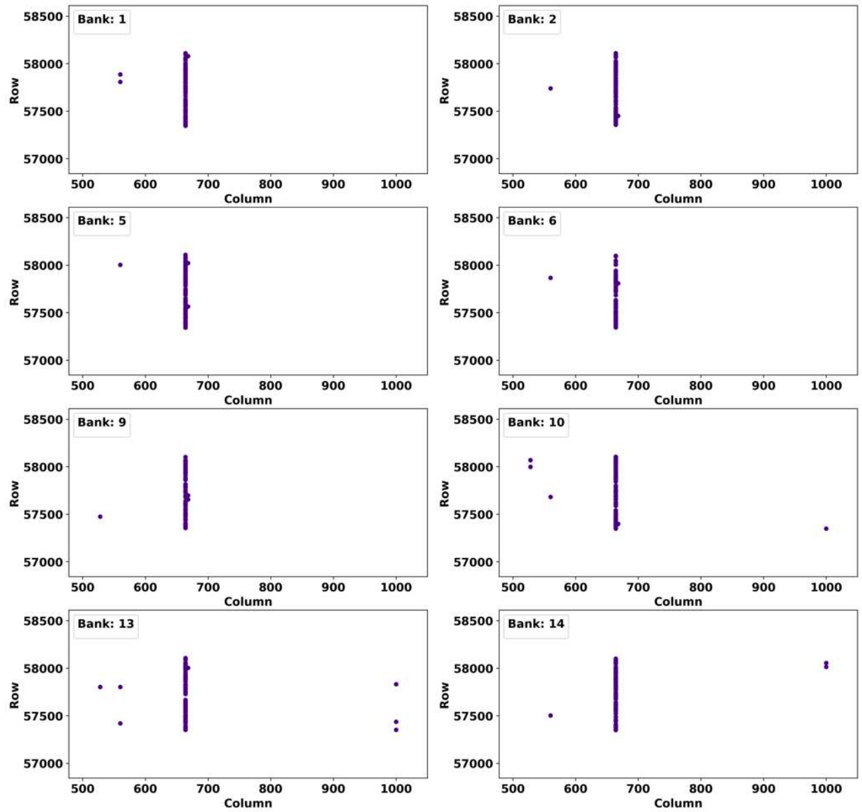
Figure 4. Unique fault modes were observed during this study (part 2).

We identify sixteen unique fault modes across the DRAM devices in our study. These fault modes are shown in Fig. 3, 4, and 5. Note that the x- and y-axis scales are different for each sub-figure to aid in the visibility of each fault mode.

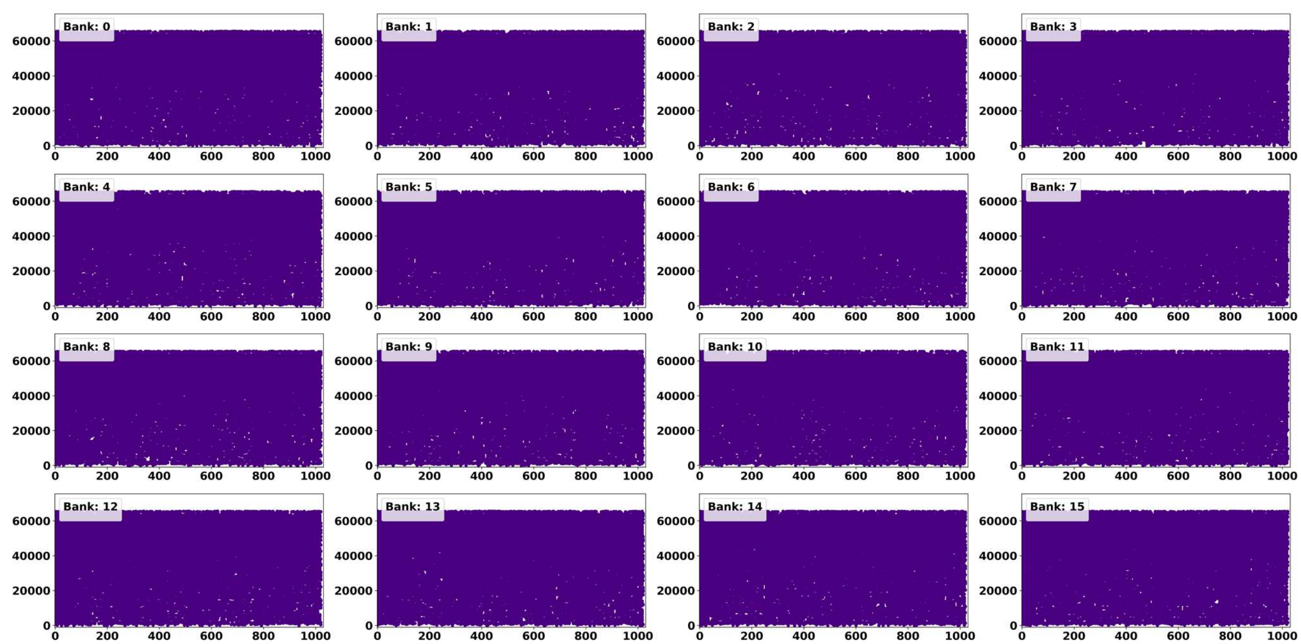
The fault modes are defined as follows:

- **Single-bit:** The fault affects a single DRAM bitcell (Fig. 3.1).
- **Single-word:** The fault affects multiple bits in a single DRAM word (not shown).
- **Single-column:** The fault affects a single DRAM column, but spans multiple rows (Fig. 3.2).
- **Two-column:** The fault affects two columns in a bank spanning multiple rows (Fig. 3.3).
- **Partial-row:** The fault affects between two and 128 columns in a row (Fig. 3.4 and 3.5).
- **Single-row:** The fault affects between 128 and 1024 columns in a row (Fig. 3.6 and 4.1).
- **Single-row-plus-single-bit:** The fault affects a single row, plus an additional bit in the same bank (Fig. 4.2). The additional bit is typically only a few rows away from the faulty row.
- **Two-row:** The fault affects exactly two rows in the same bank (Fig. 4.3). The rows are typically close together but non-adjacent.
- **Consecutive-row:** The fault affects 4 or 8 consecutive rows in a single bank (Fig. 4.4).
- **Cluster-row:** The fault affects multiple clusters of rows in a bank (Fig. 4.5).
- **Single-bank:** The fault affects multiple rows in a bank (more than a quarter of all rows) and does not fall into one of the preceding categories (Fig. 4.6).
- **Quarter-device:** The fault affects four banks in a device (Fig. 5.1). These faults typically affect only one or a handful of rows per bank.
- **Half-device:** The fault affects between five and eight banks in a device (Fig. 5.2). Similarly, to the quarter-device faults, these faults typically affect only a small fraction of each bank. Unlike quarter-device faults, the faults typically are not confined to a small number of rows in each bank.
- **Full-device:** The fault affects between nine and sixteen banks in a device (Fig. 5.3). 75% and 60%





(1) Quarter-device (2) Half-device



(3) Full-device

Figure 5. Unique fault modes were observed during this study (part 3).

TABLE I. THE PERCENTAGE OF FAULT MODES AND TYPES.

Fault Modes	Total Faults		Fault Type					
	Vendor A	Vendor B	Vendor A			Vendor B		
			Likely Transient	Intermittent	Potentially Permanent	Likely Transient	Intermittent	Potentially Permanent
Single-bit	54.64%	55.48%	46.4%	35.4%	18.2%	44.4%	35.8%	19.8%
Single-word	0.65%	0%	80.0%	20.0%	0%	0%	0%	0%
Single-column	4.27%	3.43%	79.8%	14.1%	6.1%	28.0%	48.0%	24.0%
Two-column	5.27%	0.41%	86.1%	10.7%	3.2%	0%	33.3%	66.7%
Partial-row	25.68%	23.01%	4.5%	84.4%	11.1%	3.0%	76.2%	20.8%
Single-row	0.52%	0%	0%	41.7%	58.3%	0%	0%	0%
Single-row-plus-single-bit	0.17%	1.78%	0%	75.0%	25.0%	23.1%	53.8%	23.1%
Two-row	0.17%	8.08%	25.0%	50.0%	25.0%	1.7%	55.9%	42.4%
Consecutive-row	0.43%	0.68%	60.0%	0.0%	40.0%	0%	80.0%	20.0%
Cluster-row	6.6%	4.8%	73.9%	8.5%	17.6%	22.9%	28.5%	48.6%
Single-bank	0.13%	0%	0%	0%	100%	0%	0%	0%
Quarter-device	0%	0.27%	0%	0%	0%	0%	50.0%	50.0%
Half-device	0.04%	0.14%	0%	0%	100%	0%	0%	100%
Full-device	0.52%	0.69%	41.7%	16.6%	41.7%	0%	60.0%	40.0%
Single-pin	0.52%	0.82%	0%	8.3%	91.7%	33.3%	16.7%	50.0%
Single-lane	0.39%	0.41%	22.2%	22.2%	55.6%	0%	0%	100%
Total	100%	100%	40.5%	43.5%	16.0%	28.2%	47.3%	24.5%

of these faults affect all sixteen banks for Vendor A and Vendor B, respectively. These faults typically affect half or more of the bits in each bank, spread across all the columns, similar to Fig. 5.3.

- **Single-pin:** The fault affects a single DQ pin but occurs across all ranks on that pin (not shown).
- **Single-lane:** The fault affects a single lane, but occurs across all ranks on that lane (not shown).

Table I shows the relative rate at which each of these fault modes occurs as a percentage of the total number of faults. The table shows that for both vendors ~55% of all DRAM faults are single-bit faults, while the remaining ~45% of all DRAM faults affect multiple bits. That is, multi-bit faults occur almost as frequently as single-bit faults in the DDR4 DRAMs under study, a marked change from previous DRAM technologies [1].

For single-column faults, 72.6% of these faults on Vendor A affect only a portion of a column, while 27.4% of these faults affect the entire column. For Vendor B, 96.3% of these faults only affect a portion of the column and 3.7% of them affect the entire column.

Table I shows that 25.68% and 23.01% of all faults are partial-row faults for Vendor A and Vendor B, respectively, or just about one-fourth of all faults for both vendors. Our results show that approximately 90% and 96% of these faults affect only two columns in the row. Most of these two-column partial-row faults affect non-adjacent columns. However, the physical layout of the DRAM can be different than the logical layout [36]; so, these bits may be physically adjacent within the DRAM.

TABLE II. THE FAULT RATE (FIT PER DRAM DEVICE) OF FAULT MODES

Fault Modes	FIT per DRAM Device	
	Vendor A	Vendor B
Single-bit	39.13	52.86
Single-word	0.49	0.02
Single-column	3.12	3.39
Two-column	3.83	0.45
Partial-row	18.45	22.08
Single-row	0.4	0.02
Single-row-plus-single-bit	0.14	1.8
Two-row	0.14	7.86
Consecutive-row	0.33	0.72
Cluster-row	4.79	4.71
Single-bank	0.11	0.02
Quarter-device	0.01	0.31
Half-device	0.04	0.17
Full-device	0.4	0.72
Single-pin	0.4	0.86
Single-lane	0.3	0.45
Total	72.08	96.44

Table I shows that single-bank faults are relatively rare: Single-bank faults are just 0.13% and 0% of all faults for Vendor A and Vendor B, respectively. However, when including two-column, single-row-plus-single-bit, two-row, consecutive-row, cluster-row, and single-bank faults as bank faults (as prior work has done), bank faults are 12.77% and 15.75% of all faults for Vendor A and Vendor B, respectively which is similar to the fault mode distribution in DDR2 and DDR3 [2, 11].

Table II shows the fault rate of each fault mode experienced by each vendor. The table shows that for most of the fault modes, Vendor B has a higher fault rate compared to Vendor A. However, for single-word, single-row, two-column, single-row, cluster-row and single-bank, Vendor A has a higher fault rate than Vendor B.

C. *Mapping Fault Modes to Physical DRAM Failure*

Mapping observed fault modes to physical failure boundaries in DRAM requires vendor and device-specific mapping information [36], which we do not have in this analysis. Based on our experience, most faults in DRAM are the result of faults in components internal to the DRAM device, such as a bit-line (BL), BL sense amplifier (BLSA), sub word-line (SWL) contacts, SWL drivers, or main word-line (MWL). Based on which component fails and its location in the DRAM device, the fault may manifest as a single-bit or multi-bit fault mode. For example, a BLSA fault shows up as a single-column fault, a SWL driver fault can impact multiple row addresses, and a MWL fault can cause a consecutive-row fault. By contrast, single-lane, single-pin and full-device faults are typically caused by a component external to the DRAM device such as DQ pins or connectors. For example, these faults may be due to poorly seated DIMMs, contaminated DIMM slots, or damage to the DIMM connectors.

D. *Fault Types*

Table I shows the percentage of each fault mode experienced by each vendor classified as likely transient, intermittent, or potentially permanent. The table shows that the ratio of intermittent faults to likely transient faults is 1.68x for vendor B and 1.07x for Vendor A, while the ratio of intermittent faults to potentially permanent faults is 1.9x for vendor B and 2.7x for Vendor A. The table shows that for both vendors, single-bit faults are approximately equally likely to be transient or intermittent; partial-row, single-row-plus-single-bit, and two-row are largely intermittent; and single-pin and single-lane are potentially permanent. For Vendor A, the majority of single-word, single-column, two-column, and cluster-row are likely transient.

E. *Time To Fail*

Fig. 6 shows the time-to-fail in days for single-bit and multi-bit faults for both vendors. The figure shows that the fault rate for both single-bit and multi-bit faults of Vendor B exhibits a clear “bathtub curve” effect: a high rate of faults in early life followed by a period with a lower, constant rate of faults. This is consistent with fault modes for Vendor B being largely non-transient. By contrast, single-bit and multi-bit faults for Vendor A show a much smaller “bathtub curve” effect: the time to fail is largely linear starting from time 0.

The difference in TTF characteristics between Vendor A and Vendor B could indicate, for example, that DRAM components for Vendor A have been through sufficient burn-in to avoid an elevated early-life fault rate, while DRAM components for Vendor B have not. This hypothesis is further bolstered by Vendor B’s relatively higher intermittent and potentially permanent fault rates.

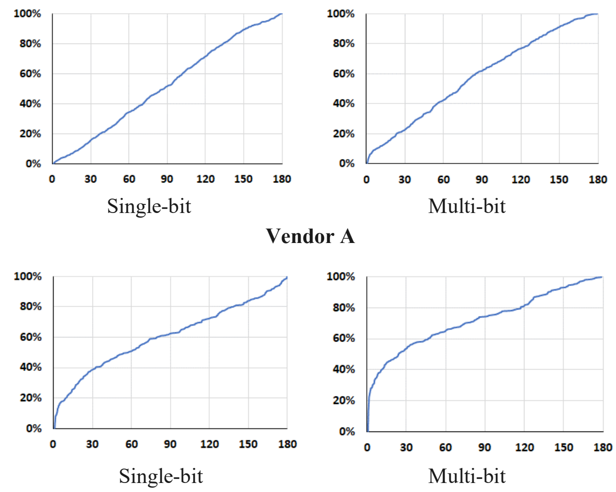


Figure 6. Time to fail in days. Single-bit and multi-bit faults for vendor A show a linear TTF, indicating a constant failure rate, while single-bit and multi-bit faults for Vendor B show a high rate of early-life failures followed by a lower steady-state rate.

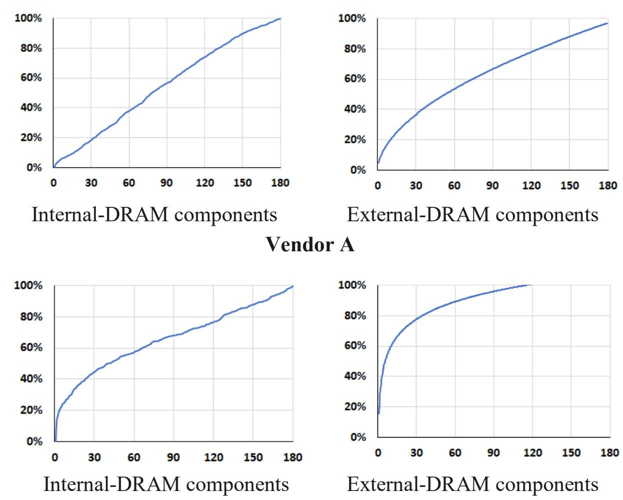


Figure 7. Time to fail in days. Components that are external to the DRAM device (External-DRAM) show a high rate of early-life failures compared to components that are internal to the DRAM device (Internal-DRAM).

We note that lanes and pins are components that are external to the DRAM device itself, while bits, rows, and columns are components internal to the DRAM device. Fig. 7 shows the time-to-fail in days for components that are external to the DRAM device (External-DRAM) and components that are internal to the DRAM device (Internal-

TABLE III. BOUNDED FAULT MODES

Bounded Fault (Per Spec and Vendor Discussion)	
Single-bit, Single-word, Single-column, Partial-row, Single-row, Single-row-plus-single-bit, Two-row and Single-pin	
Inherently Bounded	
DDR4	Two-column, Consecutive-row, Cluster-row, Single-bank, Quarter-device, Half-device, Full-device with 1DQ and 2DQ (Adjacent)
DDR5 10×4	Two-column, Consecutive-row, Cluster-row, Single-bank, Quarter-device, Half-device, Full-device with 1DQ and 2DQ (Adjacent)
DDR5 9×4	Two-column, Consecutive-row, Cluster-row, Single-bank, Quarter-device, Half-device, Full-device with 1DQ

TABLE IV. UNBOUNDED FAULT MODES

Unbounded	
DDR4	Two-column, Consecutive-row, Cluster-row, Single-bank, Quarter-device, Half-device, Full-device with 2DQ (non-adjacent), 3DQ and 4DQ
DDR5 10×4	Two-column, Consecutive-row, Cluster-row, Single-bank, Quarter-device, Half-device, Full-device with 2DQ (non-adjacent), 3DQ and 4DQ
DDR5 9×4	Two-column, Consecutive-row, Cluster-row, Single-bank, Quarter-device, Half-device, Full-device with 2DQ, 3DQ and 4DQ

DRAM) for both vendors. Results show that components that are external to the DRAM device show a high rate of early-life failures (failures within the first 30 days of deployment) compared to components that are internal to the DRAM device. The difference in TTF characteristics could indicate, for example, that DRAM-internal components have been through sufficient burn-in to avoid an elevated early-life fault rate, while DRAM-external components have not. Our results also show that full-device faults have a similar characteristic to the DRAM-external faults, this could also indicate that full-device faults are largely caused by a component external to the DRAM device itself (e.g., on the DIMM) as opposed to a component inside the DRAM device.

VIII. CASE STUDY: COMPARING ERROR-CORRECTING CODES

In this section, we use the data presented in the previous sections to compare the uncorrected memory fault rate of three ECC schemes. The three ECCs we compare are: single error correct-double error detect (SEC-DED) ECC; chipkill ECC (the ability to correct any error from a single DRAM device); and bounded fault ECC (the ability to correct any error from a bounded fault in a single DRAM device) [8]. Our results in Section VIII.C show that using SEC-DED ECC (instead of using chipkill) increases the fraction of faults that lead to an uncorrectable error. As shown in Table I, ~45% of faults affect multiple bits and cannot be corrected by SEC-DED ECC. As a result, the rate of faults that cause uncorrectable errors (i.e., UE faults) when using SEC-DED ECC is 32.95 and 43.58 FIT per DRAM device for Vendor A and Vendor B, respectively. Moreover, our results show that using bounded fault ECC increases the UE fault rate by up to 5.71 and 4.97 FIT per DRAM device for Vendor A and Vendor B, respectively, compared to using chipkill.

A. Bounded Fault ECCs

Bounded Fault (BF) is a new architecture for DDR5 where DRAM devices guarantee that a set of fault modes are confined to half the data pins of a DRAM device [8]. This allows a system designer to implement an ECC that guarantees correction of these *bounded* faults with lower overhead than a chipkill ECC.

For example, a chipkill ECC code in DDR4 requires the use of sixteen DRAM devices for data and two DRAM devices for check bits, consuming all available bits in the memory channel [37]. A "bounded fault ECC" can be implemented using fewer check bits, freeing up bits in memory for non-ECC uses.

In DDR5 [9], the ECC overhead for chipkill increases due to the use of a half-wide data channel compared to DDR4: a DDR5 x4 DIMM has eight x4 DRAM devices for data and two x4 DRAM devices for check bits on each 40-bit sub-channel (a "10×4" configuration). A bounded fault ECC can free up some of the check bits for non-ECC uses or can reduce cost by eliminating one of the DRAM devices on each sub-channel (a "9×4" configuration).

The concept of DRAM fault bounding is used in HBM3 to improve reliability, availability, and serviceability (RAS) over HBM2 by allowing certain multi-bit fault modes to be correctable [38].

B. Methodology

In this study, we project UE fault rate of: a chipkill ECC on DDR4 18×4 DIMM; a SED-DED ECC on a DDR4 18×4 DIMM; a bounded fault ECC on a DDR4 18×4 DIMM; a chipkill ECC on a DDR5 10×4 DIMM; a bounded fault ECC on a DDR5 10×4 DIMM, and a bounded fault ECC on a DDR5 9×4 DIMM. We assume an unbounded fault causes exactly one uncorrectable fault before the DIMM is repaired or replaced. This study does not consider the effect of DDR5's on-die ECC as it largely does not affect multi-bit faults, which dominate the fault rates [9]. Note that DDR5's on-die ECC was introduced to combat an expected increase in the single bit fault rate due to DRAM scaling, but the on-die ECC by itself does not change a device's fault rate [38].

We determine which faults are bounded based on the DDR5 specification and map the specified fault modes to the fault modes observed in our study. We also assume some faults are "inherently" bounded to a set of DQs (e.g., single-pin faults). Tables III and IV show the list of fault modes observed in our study which we consider bounded and unbounded, respectively, for each memory channel configuration. Note that some faults that appear unbounded (e.g., full-device faults) are considered *inherently bounded* if they only affect 1 or 2 DQs in our data depending on DIMM configurations.

C. Results

Table V shows the UE fault rate of each DIMM configuration and ECC code. DDR4 18×4 with chipkill ECC and DDR5 10×4 with chipkill ECC have 0 UE fault rate because all faults contained to a single DRAM device are correctable. In practice, chipkill ECC will still experience uncorrectable errors; however, these will all be faults that affect multiple DRAM devices.

By contrast, the DDR4 18×4 with bounded fault ECC, DDR5 10×4 with bounded fault ECC and DDR5 9×4 with bounded fault ECC increase the UE fault rate relative to DDR5 10×4 chipkill by up to 5.71 and 4.97 FIT per DRAM device, respectively for Vendor A and Vendor B. This FIT rate increase is due to the large fraction of unbounded faults observed in our study, in particular two-column and cluster-row faults. In our experience, this is a reliability degradation for a server system that may have hundreds or thousands of DRAM devices.

Fig. 8 shows the time-to-fail in days for unbounded DDR4 faults for both vendors using the approach described in Section VI.D. The figure shows that for Vendor A, the time-to-fail is linear with time, similar to the time-to-fail for multi-bit faults or DRAM-internal components described in Section VII.E. This indicates that for Vendor A, a bounded fault ECC is likely to have a constant uncorrected fault rate over the first 180 days of deployment.

By contrast, Fig. 8 shows a high rate of early-life failures for unbounded DDR4 faults for Vendor B. Specifically, this figure shows that the DDR4 with bounded fault ECC in this study experienced 60% of the uncorrected faults that occurred in the first 180 days of a node's deployment occurred within the first ten days.

TABLE V. RATE OF FAULTS IN A SINGLE DRAM DEVICE THAT CAUSE UNCORRECTABLE ERRORS (I.E., UE FAULTS)

ECC Schemes	ECC	FIT/DRAM Device	
		Vendor A	Vendor B
DDR4 (18×4)	Chipkill	0	0
	SEC-DED	32.95	43.58
	Bounded Fault	5.1	4.44
DDR5 (10×4)	Chipkill	0	0
	Bounded Fault	5.71	4.97
DDR5 (9×4)	Bounded Fault	5.35	4.58

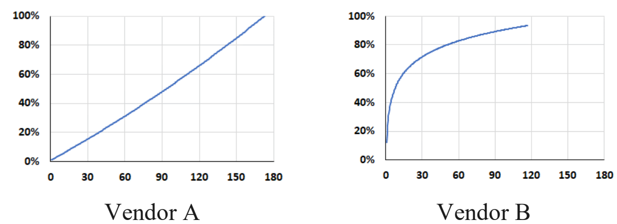


Figure 8. Time to Fail in days for DDR4 unbounded faults for Vendor A and Vendor B.

IX. CONCLUSIONS

This paper presented a detailed study of DDR4 DRAM memory faults from a large fleet of commodity servers. We identified several DRAM fault characteristics to give server architects some insights into the efficacy of existing reliability features and guidance on the design of future systems.

We believe that our findings have several implications for the architecture and design of future systems: ~45% of all faults in our DDR4 data set affect multiple bits, indicating a need for strong ECCs; a large variety of fault modes are present in DDR4 DRAM channels, indicating continued memory reliability challenges; and that chipkill continues to be effective in reducing the rate of faults that cause uncorrectable errors in DDR4 and likely DDR5.

Increasing memory reliability is a shared problem: DRAM vendors can design more resilient DRAM devices by understanding faults in current devices; processor vendors can implement a range of ECCs targeting different reliability levels; system designers can choose which ECCs to use based on reliability targets for their system; and the industry (e.g., JEDEC) can influence future DRAM architectures. Public field studies have been used to guide all of these activities in the past [38]. The goal of our study is to provide recent data to enable these actions for future DRAM, processor, and system designs.

Overall, our hope is that this work will help to influence the industry to continue improving DRAM reliability to ensure that DRAM technologies continue to be appropriate for mission-critical server applications.

X. ACKNOWLEDGEMENTS

 This paper reflects collaborative work between the authors. This paper does not express nor imply an official statement or any viewpoints of Advanced Micro Devices, Inc. (AMD).

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